

Population, Census Data, and Sample Space: Concepts and Examples

A population is the complete set of individuals or items that share a common characteristic and about which information is required for a statistical investigation. It represents the entire group of interest, such as all students in a university, all households in a city, or all products produced by a factory. Census data refers to information collected from every member of a population, providing comprehensive and accurate empirical details used for analysis and planning. In contrast, a sample space is a theoretical concept in probability that lists all possible outcomes of a random experiment, with each and every possible outcome included only once. For example, the sample space for a coin toss is $\Omega = \{\text{Head}, \text{Tail}\}$. Unlike census data, which is real and observed, the sample space is abstract and serves as a foundation for modeling uncertainty and calculating probabilities.

In the context of rolling a die, the population may refer to all the actual rolls observed in an experiment, for example, 50 rolls made during a game. If data is collected from every one of these rolls, these complete records serve as census data, since they capture information from the entire population of observed outcomes. The sample space, however, remains the theoretical set of all possible outcomes that each roll could produce, given by $\Omega = \{1, 2, 3, 4, 5, 6\}$. While the population and its census data deal with real items and observed results, the sample space represents the complete list of potential outcomes used to model uncertainty and calculate probabilities.

Boolean Algebra (or just Algebra) and σ -Algebra (Sigma-Algebra)

A Boolean algebra, say $\mathcal{A}$, on a sample space Ω is a collection of subsets (called events) of Ω that satisfies three properties:

- i. The sample space Ω itself is in the collection, i.e., $\Omega \in \mathcal{A}$.
- ii. The collection is closed under complementation: if A is in the collection, then A^c is also in the collection, i.e., if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$.
- iii. The collection is closed under union, i.e., if $A_1 \in \mathcal{A}$ and $A_2 \in \mathcal{A}$, then $A_1 \cup A_2 \in \mathcal{A}$.

A sigma-algebra differs from an algebra in that the above (iii) property is extended to countable unions, i.e., if $A_1, A_2, A_3, \dots, A_i \dots \in \mathcal{A}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$.

Probability Function

A probability function $P[\cdot]$ is a set function with domain $\mathcal{A}$ (an algebra of events or sigma-algebra) and counterdomain the interval $[0, 1]$ which satisfies the following axioms:

- i. Non-negativity: For every event $A \in \mathcal{A}$,
 $P[A] \geq 0$.

- ii. Normalization: The probability of the sure event (the sample space Ω) is 1, i.e., $P[\Omega]=1$.
- iii. Countable additivity (for mutually exclusive events): For any countable sequence of mutually exclusive events $A_1, A_2, A_3, \dots, A_i \dots \in \mathcal{A}$ such that $A_i \cap A_j = \emptyset$ for $i \neq j$ and if the union $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$, then

$$P\left[\bigcup_{i=1}^{\infty} A_i\right] = \sum_{i=1}^{\infty} P[A_i].$$

Note that the axioms are reasonable assumptions generally accepted as true. Irrespective of whether probabilities are interpreted a priori, posteriori, or subjectively, these axioms should be universally agreed upon; otherwise, the concept of probability would lose its mathematical consistency and reliability.

The above definition of probability is a mathematical definition; it tells us which set functions may be considered probability functions. It does not tell us what value the probability function $P[\cdot]$ assigns to any specific event A . To assign actual numerical probabilities, one must introduce additional assumptions or construct an appropriate model of the random experiment being studied, for example, by specifying physical symmetries, appealing to empirical relative frequencies, or adopting an underlying theoretical distribution. Only after such modeling can the probability function be fully determined for the situation at hand.

Limitations of Algebras

An algebra (or σ -algebra) of events $\mathcal{A}$ cannot generally contain all subsets of a sample space Ω because:

- a. Non-measurable sets exist when the sample space is uncountable.
- b. Assigning probabilities to all subsets leads to contradictions with additivity.
- c. The collection of *all* subsets is too large and contains "pathological" sets (that break the axioms of probability or measure theory).
- d. Probability theory requires sets to be measurable, and not all subsets are measurable.

Thus, an algebra must be a carefully chosen collection of subsets (events) that ensures the probability function is well-defined, consistent with axioms, and usable.

Probability Space

A probability space is a mathematical framework used to model random experiments. It is defined as a triple $(\Omega, \mathcal{A}, P[\cdot])$, where:

- i. Ω is the sample space, the set of all possible outcomes of the experiment.

- ii. $\mathcal{A}$ is a σ -algebra (sigma-algebra) of subsets of Ω , representing all measurable events.
- iii. $P[\cdot]$ is a probability function that assigns a probability $P[A]$ to each event $A \in \mathcal{A}$, satisfying the axioms of probability (non-negativity, normalization, and countable additivity).

The probability space serves as the foundation for probability theory. It allows for precise calculation of probabilities, modeling of random processes, and the development of further concepts such as conditional probability, independence, and random variables.

Random Variable

A random variable, denoted by $X(\cdot)$ or simply X , for a given probability space $(\Omega, \mathcal{A}, P[\cdot])$, is a function with domain Ω and codomain the real line $\mathcal{R}$. The function $X(\cdot)$ must satisfy the condition that for every real number r , the set $A_r = \{\omega: X(\omega) \leq r\}$ belongs to $\mathcal{A}$, meaning it represents an event. In this way, a random variable provides a way to describe events numerically.

Essentially, a random variable allows us to translate the abstract events of an experiment into numeric quantities that can be analyzed statistically.

Example:

Suppose

$\Omega = \{\text{HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT, THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH, TTTT}\}$ represents the set of outcomes from flipping a coin four times. A random variable X could assign to each outcome the number of heads in that sequence. In this case, instead of needing all $2^{16} = 65,536$ subsets of Ω to describe events, we only need 5 subsets corresponding to $X=0, 1, 2, 3, 4$ heads.

Thus, a random variable uses Ω and $\mathcal{A}$ to define events in terms of real numbers. Using the random variable, one can assign probabilities to events through functions such as the cumulative distribution function (CDF), probability mass function (PMF), or probability density function (PDF). This approach allows us to model a random experiment by making assumptions about the random variable and its distribution, often bypassing the need to explicitly describe the full probability space.

In essence, a probability space provides the foundational structure. A random variable maps outcomes and events from this space to real numbers, the CDF summarizes the distribution of these values, and the PDF (or PMF) quantifies probabilities for individual values or intervals. Together, they form the core tools for modeling and analyzing uncertainty.

Connecting Binomial and Poisson Distributions: Theory and Applications

1. From Discrete to Continuous Exponential Growth and the Poisson Limit

Suppose a college initially has $S_0=1000$ students. If each student, on average, attracts $\lambda=5$ new students per year, then after 1 year the total number of students will be

$$S_0 + S_0\lambda = S_0(1+\lambda) = 1000(1+5).$$

Now consider that new admissions occur continuously; that is, the year is divided into many small intervals (say, per day). If admissions occur **independently** in non-overlapping subintervals of time, the growth process approaches exponential growth. With $n=365$ days in a year, the maximum possible total number of students under continuous compounding is approximately

$$S_0 \left(1 + \frac{\lambda}{365}\right)^{365} = S_0 \left(1 + \frac{\lambda}{n}\right)^n \approx S_0 e^\lambda = S_0 e^5, \text{ as } n \rightarrow \infty.$$

Since

$$\begin{aligned} e^\lambda &\approx \left(1 + \frac{\lambda}{n}\right)^n = {}^nC_0 \left(\frac{\lambda}{n}\right)^0 + {}^nC_1 \left(\frac{\lambda}{n}\right)^1 + {}^nC_2 \left(\frac{\lambda}{n}\right)^2 + {}^nC_3 \left(\frac{\lambda}{n}\right)^3 + \dots \\ &= \sum_{x=0}^n \frac{n!}{x!(n-x)!} \frac{\lambda^x}{n^x} = \sum_{x=0}^n \frac{n(n-1)(n-2)\dots(n-(x-1))(n-x)!}{x!(n-x)!} \frac{\lambda^x}{n^x} \\ &= \sum_{x=0}^n \frac{1 \cdot \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{x-1}{n}\right)}{x!} \lambda^x = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}, \text{ as } n \rightarrow \infty \end{aligned}$$

we can equivalently express the maximum possible total as

$$S_0 e^\lambda = S_0 \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$$

Thus, when student recruitment is viewed as a rare event occurring daily or continuously (i.e., the number of "trials" becomes very large), the number of students a given student attracts, taking only non-negative integer values $0, 1, 2, 3, \dots, \infty$, follows a **Poisson distribution**:

$$P(X=x) = \frac{S_0 \frac{\lambda^x}{x!}}{S_0 e^\lambda} = \frac{\lambda^x}{x!} \frac{1}{e^\lambda} = \frac{\lambda^x e^{-\lambda}}{x!} \quad x = 0, 1, 2, \dots, \infty$$

2. Modeling Rare Events in Fixed Intervals: From Binomial to Poisson

Suppose time is fixed to one unit (cross-section data) and events occur at a rate λ . Divide the interval $(0, 1)$ into n subintervals. The probability of exactly one event occurring in any small subinterval is approximately $p = \frac{\lambda}{n}$, and the probability of no event is approximately $1 - p = 1 - \frac{\lambda}{n}$. Each subinterval constitutes an **independent** Bernoulli trial. Therefore, the probability of observing exactly x events in the n trials is given by Binomial distribution

$$P(x = x) = {}^nC_x p^x (1 - p)^{n-x} = {}^nC_x \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

which *approximates* the probability that x events occur in the time interval $(0, 1)$.

Taking the limit as the number of subintervals increases, $n \rightarrow \infty$, and the probability of an event in each subinterval decreases, $p (= \frac{\lambda}{n}) \rightarrow 0$, yields the exact expression of Poisson distribution

$$\begin{aligned} {}^nC_x \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} &= \frac{n(1-n)(1-2)\dots(n-(x-1))}{x!} \cdot \frac{1}{n^x} \lambda^x \cdot \frac{\left(1 - \frac{\lambda}{n}\right)^n}{\left(1 - \frac{\lambda}{n}\right)^x} \\ &= \frac{1 \cdot \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{x-1}{n}\right)}{x!} \cdot \lambda^x \frac{\left(1 - \frac{\lambda}{n}\right)^n}{\left(1 - \frac{\lambda}{n}\right)^x} = \frac{\lambda^x}{x!} e^{-\lambda} \text{ as } n \rightarrow \infty \end{aligned}$$

Thus, when selecting a sample of size n from a population to assess a characteristic, the probability of success remains constant only if sampling is done **with replacement**; in this case, the number of successes in n *independent* trials follows a **binomial distribution**. When n becomes large and the success probability p becomes small such that $np = \lambda$ remains fixed, the binomial distribution can be approximated by the Poisson distribution. In this limiting case, the Poisson distribution provides the exact model for the number of rare events occurring in a fixed interval of time or space.

A binomial distribution models the number of successes in n *independent* Bernoulli trials with success probability p , whereas a Poisson distribution models the number of events occurring in a **fixed interval or space** when events occur *independently* with an average rate λ .

Derivation of the Probability Density Function for the Exponential Distribution

To derive the Probability Density Function (PDF) of the exponential distribution, we begin with the Poisson distribution. The Poisson distribution models the probabilities associated with counting the number of occurrences of a given happening within a fixed interval of time or space. Its probability mass function (PMF) is given by:

$$P(k; \lambda) = \frac{e^{-\lambda} (\lambda)^k}{k!}; \quad k = 0, 1, \dots, \infty$$

When we extend this to a continuous setting, treating time as a continuous variable, we arrive at what is known as a Poisson process. A Poisson process models the occurrence of happenings over continuous time rather than at fixed points or intervals - for example, traffic accidents on a highway or incoming phone calls. It assumes that happenings occur independently and at a constant average rate, with the expected number of happenings being *proportional* to the length of the time interval. Mathematically, it's represented as:

$$P(k, t; \lambda) = \frac{e^{-\lambda t} (\lambda t)^k}{k!}; \quad k = 0, 1, \dots, \infty$$

Where:

- $P(k, t)$ is the probability of observing k happenings over the continuous time interval $[0, t]$.
- λ is the rate parameter, representing the average rate of the happening occurrences per unit time.
- t is the length of the continuous time interval.

Now, consider the probability of observing the occurrence of the happening within the interval $[0, t]$. Any count other than zero implies that the happening has occurred. Suppose that one of these happenings has just occurred; we then ask: what is the distribution of the waiting time, denoted by X , until the next happening occurs? The cumulative distribution function (CDF) of the waiting time can therefore be obtained by summing the relevant probabilities and expressing them as a function of the *waiting time* x :

$$F_X(x) = 1 - e^{-\lambda x}; \quad x > 0.$$

Now, differentiating this CDF with respect to time x , we obtain the PDF of the exponential distribution that describes the *probability distribution of the waiting time between successive happenings* in a Poisson process:

$$f_x(x) = \frac{dF(x)}{dx} = \begin{cases} \lambda e^{-\lambda x}; & x > 0, \lambda > 0 \\ 0, & \text{otherwise.} \end{cases}$$

This function describes the probability density of the exponential distribution, where λ represents the average rate of the happening occurrences per unit time. In other words, it describes the probability of waiting a certain amount of continuous time before a happening occur, assuming the happening is occurring randomly and **independently** at a constant average rate.

Note:

- i. The PMF of geometric distribution is given by

$$f(x; q) = \begin{cases} pq^x = p(1-p)^x & \text{for } 0 < q \leq 1, \quad x = 0, 1, 2, \dots, \infty \\ 0 & \text{otherwise.} \end{cases}$$

When $p \rightarrow 1$, $(1-p) = e^{-p}$, and we the PMF as

$$f(x; q) = \begin{cases} pe^{-px} & \text{for } 0 < q \leq 1, \quad x = 0, 1, 2, \dots, \infty \\ 0 & \text{otherwise.} \end{cases}$$

That is, when event 1 occurs with a probability p , we vary the total number of trials or items, n , to obtain different probabilities for different values of n .

However, when event 1 occurs with an average value of $\lambda=np$, we vary the time, t , to determine the probabilities for different values of t . In this case, the distribution transitions to an exponential distribution.

In other words, the exponential distribution can be considered the continuous equivalent of the geometric distribution, which models the number of Bernoulli trials required for a discrete process to transition to a new state. In comparison, the exponential distribution represents the time required for a continuous process to undergo a state change.

Example-1: In practical scenarios, the assumption of a constant rate (or probability per unit time) is seldom met. For instance, the frequency of incoming phone calls varies throughout the day. However, if we consider a specific time period during which the rate remains approximately constant—such as between 2 p.m. and 4 p.m. on workdays—the exponential distribution can serve as a reasonable model for predicting the time until the next phone call.

Example-2: Exponential variables are also applicable for modeling situations where events occur at a constant probability per unit measure. Examples include the distance between mutations along a DNA strand or the spacing between road kill occurrences on a specific roadway.

Why Zero Probability Appears Excluded in the Derivation of the Exponential Distribution from a Poisson Process

When deriving the exponential distribution from a Poisson process, the probability of "0" is not arbitrarily excluded; rather, it plays a crucial role through $P(N(t)=0)$, the probability that no happenings occur in the interval $[0,t]$, which equals $e^{-\lambda t}$ (which also depends on time) and defines the survival function of the waiting time. The apparent exclusion arises only when considering $P(X=0)$, the probability that the *waiting time* is exactly zero. This probability is zero because the waiting time is a continuous random variable, and point probabilities in continuous time vanish. Intuitively, while a happening cannot occur instantaneously at time zero with positive probability, it may occur within an arbitrarily small interval immediately after zero. Thus, "0" happenings are essential in the derivation, but an exact zero waiting time has probability zero due to the continuity of time.